

# TFPT — Appendix H: the horizon unit system

One seam constant  $c_3 = \frac{1}{8\pi}$  as the universal horizon thermal code  
(Hawking, de Sitter, Unruh, Page, scrambling — *standard physics in seam units, not new gravity*)

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## What this document is about

A change of bookkeeping, not new gravitational physics: if gravity is the geometry-channel readout of the seam, then *all* horizons read the same boundary constant  $c_3 = \frac{1}{8\pi}$ . This note collects the readouts — Hawking, de Sitter and Unruh temperature, black-hole thermodynamics, the Page time, scrambling, the Nariai bound,  $v_{\text{GW}} = c$  and cosmic birefringence — in seam units, with two genuine compiler fingerprints ( $1920 = |W(D_5)|$ ,  $|\mu_4| = 4$ ).

## The TFPT document set — what is treated where

**Plain language:** TFPT is a small discrete compiler. Two inputs — the seam constant  $c_3 = \frac{1}{8\pi}$  and the carrier rank  $g_{\text{car}} = 5$  — build  $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$  and read off the Standard Model, the constants and the scale grammar. The development is **five short documents plus this appendix**, best read in order:

1. **introduction** — reading guide, compiler closure, paper-by-paper comparison, predictions, the dependency DAG and proof ledger.
2. **tfpt\_1\_architecture\_e8** — the two axioms, the derivation map, the EM fixed point  $\alpha^{-1}$ , the  $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$  construction.
3. **tfpt\_2\_standard\_model** — the SM in one  $\varphi_0$ -ladder formula, flavor from parabolic transport, the worked closures, and gravity/QG as the seam response.
4. **tfpt\_3\_e8\_audit\_bootstrap** — the seven  $E_8$  slices as an audit raster, the cascade bridge, the Möbius bootstrap.
5. **tfpt\_4\_frontier** — honest status of  $\eta_B$ ,  $m_p/m_e$ , Koide, dark matter, full QG.
- H. **tfpt\_horizon\_readouts** — **Appendix H**: one seam constant as the universal horizon thermal code (a unit-system reframe, *not* a peer paper, *not* new physics).

**You are reading Appendix H** — a notation/unit reframing; all new-physics content lives in documents 1–4.

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### Scope and honest typing

If gravity is the geometry-channel readout of the seam (companion note “Seam response grammar”), then horizons are not separate miracles — they all read the *same* boundary normaliser  $c_3 = 1/(8\pi)$ . This note collects the readouts. Markers: **[P]** standard horizon physics rewritten in TFPT/seam units (no new claim), **[I]** an exact compiler connection (audit-level), **[N]** a numerical readout (reproduced), **[A]** a search target, not a result. Nothing here is sold as new gravitational physics; it is a *change of bookkeeping* that exposes shared structure.

### Unit conventions used throughout

Three unit systems appear; the seam factor  $c_3 = 1/(8\pi)$  is the same number in all of them, only the carried dimensionful constants change:

| system         | convention                                                            | Hawking temperature                                                    |
|----------------|-----------------------------------------------------------------------|------------------------------------------------------------------------|
| Planck         | $G=\hbar=c=k_B=1$                                                     | $T_H = c_3/M = 1/(8\pi M)$                                             |
| reduced Planck | $\bar{M}_{\text{Pl}} = M_{\text{Pl}}/\sqrt{8\pi},$<br>$\hbar=c=k_B=1$ | $T_H = c_3 \bar{M}_{\text{Pl}}^2/M$                                    |
| SI             | restore $G, \hbar, c, k_B$                                            | $T_H = \frac{\hbar c^3}{8\pi G k_B M} = c_3 \frac{\hbar c^3}{G k_B M}$ |

All boxed formulas below are in Planck units unless stated; the seam constant is dimensionless.

*Standard references for the formulas reframed here:* Hawking temperature and radiation (Hawking 1975); Bekenstein–Hawking entropy (Bekenstein 1973, Hawking 1975); Page time (Page 1993); fast scrambling  $t_{\text{scr}} \sim \beta \log S$  (Sekino–Susskind 2008, Maldacena–Shenker–Stanford 2016); Nariai bound (Nariai 1950); de Sitter thermodynamics (Gibbons–Hawking 1977). *Nothing below is a new derivation of these* — only a rewriting in the seam unit  $c_3$ .

## 1 The seam constant is the universal horizon temperature factor

Every Killing horizon has  $T = \hbar\kappa/(2\pi c k_B)$ . Since

$$\boxed{\frac{1}{2\pi} = 4c_3, \quad \frac{1}{8\pi} = c_3} \quad \text{[I]},$$

the universal factor *is* the seam constant:  $T_{\text{hor}} = 4c_3 \hbar\kappa/(c k_B)$ . Black holes, de Sitter and Unruh therefore share *one* thermal grammar. **[P]** [\[verification/v8\\_horizon.py\]](#)

## 2 Schwarzschild thermodynamics in four $c_3$ -lines

For a non-rotating, uncharged hole in Planck units ( $G=\hbar=c=k_B=1$ ):

$$\boxed{T_H = \frac{c_3}{M}, \quad S_{\text{BH}} = \frac{M^2}{2c_3}, \quad P_H = \frac{c_3}{1920 M^2}, \quad \tau_{\text{evap}} = \frac{640}{c_3} M^3} \quad \text{[P]}.$$

Two of the constants are compiler numbers, not coincidences:

- the Hawking temperature factor is  $c_3 = 1/(8\pi)$  — the same P1 boundary normaliser; [I]
- the radiated-power denominator is  $1920 = |W(D_5)|$ , the Weyl-group order of the  $D_5$  carrier. [I]

*Scope of  $P_H$* : this is the idealised massless single-channel blackbody normalisation. Greybody factors and the radiated species content shift the phenomenological evaporation coefficient; the  $c_3$ -form is the convention statement, not a general phenomenological prediction.

| object                                                  | $T_H$                           | lifetime / entropy                                                                                 |
|---------------------------------------------------------|---------------------------------|----------------------------------------------------------------------------------------------------|
| solar-mass hole                                         | $6.17 \times 10^{-8} \text{ K}$ | $\tau_{\text{evap}} \approx 2.1 \times 10^{67} \text{ yr};$<br>$S/k_B \approx 1.05 \times 10^{77}$ |
| PBH evaporating now ( $1.7 \times 10^{11} \text{ kg}$ ) | $k_B T \approx 61 \text{ MeV}$  | relevant for $\gamma/\nu$ PBH signatures                                                           |

### 3 Page time and scrambling

Since  $S_{BH} \propto M^2$  and  $\tau \propto M^3$ , the Page point ( $M \rightarrow M/\sqrt{2}$ ) is

$$t_{\text{Page}} = \left(1 - \frac{1}{2\sqrt{2}}\right) \tau_{\text{evap}} \approx 0.6464 \tau_{\text{evap}} \quad [\text{P}].$$

In TFPT the Page curve is *boundary recoverability*: the physical sector is OS-reconstructed and gapped, and the recovered mutual information decays at the transfer rate,  $I(A:C|B)_n \sim \lambda_2^n = (2/3)^{6n}$ . **This is the same operator that builds the Standard Model**: the boundary transport spectrum is  $\{1, (2/3)^6, (1/3)^6\}$  and its sub-leading eigenvalue  $\lambda_2 = (2/3)^6$  is exactly the flavor-gap eigenvalue of Paper 2 ( $\Delta_{\text{gap}} = -\log(2/3)^6 = 6 \log \frac{3}{2}$ ). One boundary transport thus governs *both* the SM flavor hierarchy and the horizon information recovery. [I] [verification/v54\_seam\_horizon\_keystones.py] The scrambling prefactor carries a  $\mu_4$  signature: with  $\beta = 1/T_H = 8\pi M$ ,  $\beta/2\pi = 4M$ , so

$$t_{\text{scr}} \sim |\mu_4| M \log S \quad (|\mu_4| = 4) \quad [\text{I}]\text{-audit.}$$

### 4 De Sitter and the Nariai bound from the $\Lambda$ closure

With the action-grammar Hubble/ $\Lambda$  readouts ( $H_0/\bar{M}_{\text{Pl}} = e^{-\alpha_\star^{-1}}/(2\pi\sqrt{\Omega_\Lambda})$ ,  $\rho_\Lambda/\bar{M}_{\text{Pl}}^4 \sim e^{-2\alpha_\star^{-1}}$ ), the asymptotic de Sitter horizon gives

$$\frac{T_{dS}}{\bar{M}_{\text{Pl}}} = 16c_3^2 e^{-\alpha_\star^{-1}}, \quad S_{dS} = \frac{e^{2\alpha_\star^{-1}}}{128c_3^4} = 32\pi^4 e^{2\alpha_\star^{-1}} \approx 3.32 \times 10^{122} \quad [\text{P}].$$

This is the area-law horizon capacity of the universe from  $(\alpha_\star^{-1}, c_3, H_\Lambda)$  alone. The maximal black hole in de Sitter (Nariai) is

$$M_N \approx 2.16 \times 10^{22} M_\odot \approx 4.3 \times 10^{52} \text{ kg}, \quad S_N = \frac{1}{3} S_{dS} \quad [\text{P}],$$

the absolute gravitational-capacity limit, not an observed structure.

### 5 The maximal black hole is the anchor: SdS in seam units

The Nariai limit is not just a capacity number — written in compiler units, the entire Schwarzschild–de Sitter (SdS) family lands on atoms that are already load-bearing elsewhere in TFPT, with no adjustable parameter on either side. [I] [verification/v101\_horizon\_anchor.py]

**Nariai = the anchor; the Koide 2/3 is the entropy bound** [I]  
 ([verification/v101\_horizon\_anchor.py])

With  $f(r) = 1 - 2M/r - \Lambda r^2/3$ , the horizon cubic at the maximal mass  $M_N = 1/(3\sqrt{\Lambda}) = 1/(N_{\text{fam}}\sqrt{\Lambda})$  is, in units  $1/\sqrt{\Lambda}$ ,

$$\boxed{t^3 - 3t + 2 = (t - 1)^2(t + 2)} \quad \text{roots } (1, 1, -2),$$

with coefficients  $(-3, 2) = (-N_{\text{fam}}, |\mathbb{Z}_2|)$  — and the root pattern is exactly the **traceless projection of the anchor**:  $a - \frac{p_1(a)}{3}\mathbf{1} = -\frac{1}{3}(1, 1, -2)$  for  $a = (1, 1, 2)$  (compare  $\chi_a = (t - 1)^2(t - 2)$ , Paper 1). Five exact companions:

- **Entropy bound = the Koide branch value.** Each Nariai horizon carries  $S_{dS}/3 = S_{dS}/N_{\text{fam}}$  and the total is  $\boxed{S_{\text{Nariai}}/S_{dS} = 2/3 = |\mathbb{Z}_2|/N_{\text{fam}}}$  (deficit exactly  $S_{dS}/3$ ). The interpolation in  $x = r_b/r_c$  is the closed form  $S_{\text{tot}}/S_{dS} = (x^2 + 1)/(x^2 + x + 1)$  — denominator =  $\Phi_3(x)$ , the  $N_{\text{fam}}$  cyclotomic — monotone from 1 to 2/3 with its minimum exactly at the merge (Fig. 2a).
- **Three-sheet conservation.** The cubic is traceless with fixed  $e_2$ , so  $\pi \sum_i r_i^2 = 6\pi/\Lambda = |\mathbb{Z}_2| S_{dS}$  for *every*  $M$ : the black hole redistributes entropy between the two physical sheets and the virtual third root; the three-sheet total never moves.
- **Koide-form quotient of the roots.**  $Q_{\text{geom}} = \sum r_i^2 / (\sum |r_i|)^2$  runs monotonically over  $[\frac{3}{8}, \frac{1}{2}]$ : pure dS gives  $\frac{1}{2} = |\mathbb{Z}_2|/|\mu_4| = \delta$  (the half-step), Nariai gives  $\frac{3}{8} = p_2(a)/e_1(a)^2$  — exactly the two nonzero  $SU(4)_1$  conformal weights (Fig. 2b).
- **The mass line is a double cover.**  $\text{disc}_r \propto (1 - 3m)(1 + 3m)$  in  $m = M\sqrt{\Lambda}$ : branch points  $m = \pm 1/N_{\text{fam}}$ , separation  $2/3 = |\mathbb{Z}_2|/N_{\text{fam}}$ , deck involution = the black-hole  $\leftrightarrow$  cosmological horizon swap — the gravitational twin of the flavor sheet  $\mathbb{Z}_2$  and of the split  $N_{\text{fam}}$ -linear cover  $(3x+2)(3x+5)$  of Paper 2 (Fig. 1).
- **Temperature lemma / orientation.**  $|\kappa_b/\kappa_c| = (2x+1)/(x(x+2)) > 1$  for  $x < 1$  (equality iff  $x = 1$ ): the black-hole sheet is *always* hotter, so evaporation flows *away* from the merge — the anchor configuration is the **repeller** and the democratic endpoint the attractor, the same orientation as the flavor flow (Koide attractor / carrier repeller, Paper 4). *Audit reading.*

Completing the seam-unit table:  $dM = c_3 \kappa dA$  (first law),  $M = 2c_3 \kappa A$  (Smarr),  $S \leq ER/(4c_3)$  (Bekenstein),  $P_H = c_3/(1920 M^2)$ , lifetime  $\tau = 5120\pi M^3 = 128 g_{\text{car}} M^3/c_3$  ( $128 = 2^7$  the dS prefactor, 7 the scalaron exponent; photon/geometric-optics conventions, typed external), Kerr  $A_{\text{ext}} = M^2/c_3$  with  $A_{\text{ext}}/A_{\text{Schw}} = 1/2 = 1/|\mathbb{Z}_2|$  and  $S_{\text{ext}} = M^2/(4c_3)$ , and the Bekenstein–Mukhanov–Hod area quantum  $4 \ln 3 = \ln 81 = \ln(N_{\text{fam}}^4) = \ln(\text{disc of the flavor cover})$  ([P] as the QNM reading). *Honest scope*: the GR identities are textbook facts; the [I] content is six independent landings on already-load-bearing atoms with zero free parameters. The “carrier in the bulk” reading (two glue chiralities = two horizon sheets, ramification = Nariai) stays [P]. No new data test ( $M_N$  today  $\sim 10^{22} M_\odot$ ): a structure surface, not a measurement surface.

**One orientation: the anchor is the stationary repeller in both sectors** [I]/[P]  
 ([verification/v102\_seam\_orientation.py])

The evaporation direction of the temperature lemma upgrades to exact variational structure on both one-dimensional moduli lines (Fig. 4). **Flavor**: the canonical relaxation  $\dot{q} =$

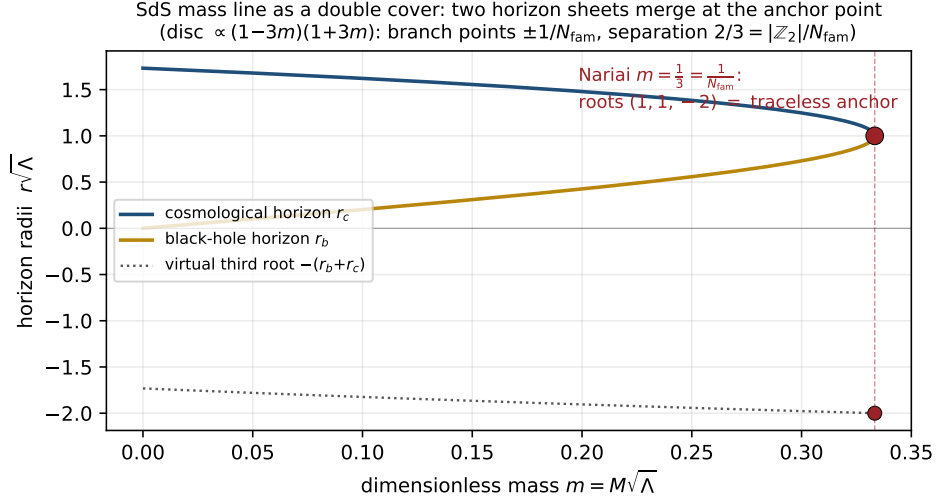


Figure 1: The SdS mass line as a double cover: black-hole and cosmological horizon sheets merge at the Nariai point  $m = 1/N_{\text{fam}}$ , where the three roots form the traceless anchor pattern  $(1, 1, -2)$ .  
[verification/v101\_horizon\_anchor.py]

$(\Delta/N_{\text{fam}})(q-2)(q-5)$  is the *gradient flow* of the cubic potential

$$V(q) = -\frac{\Delta}{N_{\text{fam}}} \left( \frac{q^3}{3} - \frac{7}{2}q^2 + 10q \right), \quad \boxed{V''(2) = +\Delta, \quad V''(5) = -\Delta}$$

— the critical points are exactly the two branch points, the stationary curvatures are  $\pm$ the transfer gap, the inflection sits at  $q = \frac{7}{2} = \text{scalaron}/2$ , and the Lyapunov rate is constant:  $d(-\ln \rho)/dt = \Delta$  exactly. **Gravity:** the SdS entropy functional has

$$\frac{d}{dx} \frac{S_{\text{tot}}}{S_{\text{dS}}} = \frac{(x-1)(x+1)}{\Phi_3(x)^2}, \quad \boxed{\left( \frac{S_{\text{tot}}}{S_{\text{dS}}} \right)''(1) = \frac{2}{9} = \frac{|\mathbb{Z}_2|}{N_{\text{fam}}^2} = \frac{2}{3} \cdot \frac{1}{3}}$$

— the Nariai/anchor point is the *unique* physical stationary point, its curvature is a grammar constant (= the product of the two Nariai entropy fractions), and the temperature lemma makes the evaporation an entropy *ascent* away from it. **Orientation theorem [I]:** in both sectors the anchor-type configuration is a stationary point of the natural functional and the dynamics flows away from it, with grammar-constant stationary curvatures. *Honest disanalogies (recorded):* the flavor attractor is itself a branch point while the gravity attractor is the smooth endpoint; the flavor functional is the potential of a [P]-typed relaxation while the gravity functional is the Bekenstein–Hawking entropy — “one variational principle of the seam” stays a *reading* [P], and no open gate is touched.

**The trisection normal form: the canonical coordinate exists [I]**  
( [verification/v103\_trisection\_normal\_form.py] )

The curvature-matching question of the previous box has a closed answer. The SdS horizon cubic is uniformized by **angle trisection**: with  $r = 2 \cos \theta$  (units  $1/\sqrt{\Lambda}$ ),  $r^3 - 3r = 2 \cos 3\theta$  *exactly*, so the cubic is  $\cos 3\theta = -3m$  — the three horizon roots are the three trisection branches (the  $\mathbb{Z}_3$  rotation = the triality deck, cf. coker  $Q = \mathbb{Z}/N_{\text{fam}}$ ). In the centered angle

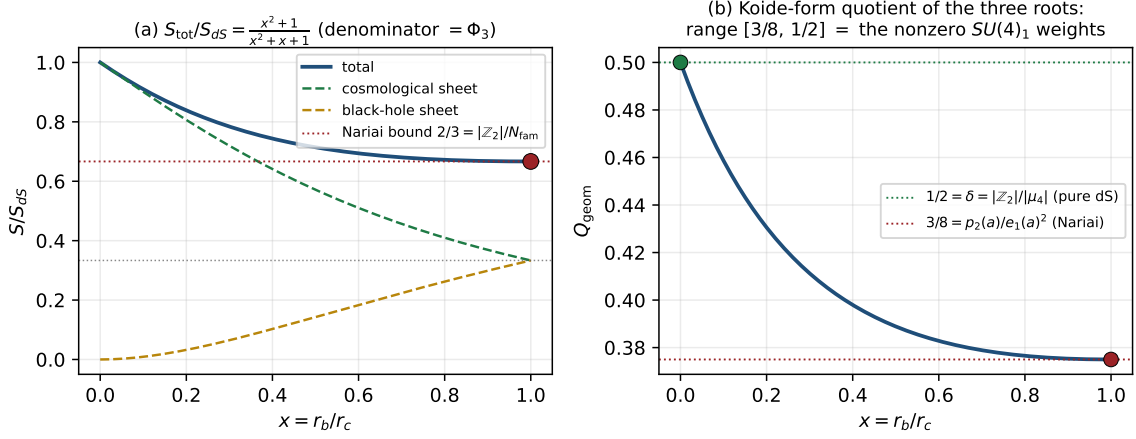


Figure 2: (a) Total SdS entropy interpolates from  $S_{dS}$  to the Nariai bound  $\frac{2}{3}S_{dS}$  via  $(x^2+1)/\Phi_3(x)$ . (b) The Koide-form quotient of the three roots runs exactly over the nonzero  $SU(4)_1$  weights  $[\frac{3}{8}, \frac{1}{2}]$ . [\[verification/v101\\_horizon\\_anchor.py\]](#)

Twin double covers: the same split  $N_{\text{fam}}$ -linear form in flavor and in classical gravity

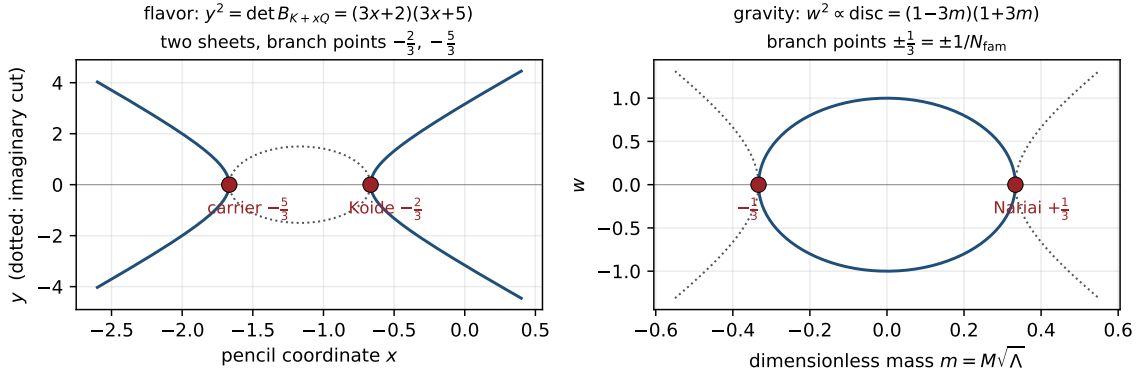


Figure 3: Twin double covers: the flavor anchor-block cover  $y^2 = (3x+2)(3x+5)$  (Paper 2) and the SdS mass-line cover  $w^2 \propto (1-3m)(1+3m)$  share the split  $N_{\text{fam}}$ -linear form with rational branch points at spine-over-family values. [\[verification/v101\\_horizon\\_anchor.py\]](#)

$\psi$  ( $\cos \varphi = -3m$ ,  $\psi = \varphi - \pi$ ; Nariai  $\psi = 0$ , pure dS  $\psi = \pi/2$ , deck  $\psi \rightarrow -\psi$ ):

$$m = \frac{\cos \psi}{N_{\text{fam}}}, \quad \boxed{\frac{S_{\text{tot}}}{S_{dS}} = \frac{4}{3} - \frac{2}{3} \cos \frac{2\psi}{3}}$$

— *one* cosine whose mean ( $|\mu_4|/N_{\text{fam}}$ ), amplitude and frequency (both  $|\mathbb{Z}_2|/N_{\text{fam}}$ ) are glue atoms over the family count (Fig. 5). Three exact consequences:

- **Canonical curvature = the Koide constant to the family power:**  $\sigma''(\psi=0) = \frac{8}{27} = (\frac{2}{3})^3 = (|\mathbb{Z}_2|/N_{\text{fam}})^{N_{\text{fam}}}$ .
- **Invariant base slope:**  $d\sigma/dm|_{\text{Nariai}} = -\frac{8}{9} = -\text{rank } E_8/N_{\text{fam}}^2$  (coordinate-invariant, cross-checked in both coordinates; generally  $-\frac{4}{3} \sin(2\psi/3)/\sin \psi$ ); dimensionful  $dS_{\text{tot}}/dM|_N = -r_N/(N_{\text{fam}}c_3)$ .
- **The bridge:** the flavor invariant is a *rate*, multiplier  $(2/3)^6 = (2/3)^{2N_{\text{fam}}}$  per transport step; the gravity invariant is a *curvature*,  $(2/3)^{N_{\text{fam}}}$  in the canonical angle — same base

One orientation: the anchor is the stationary repeller in both sectors

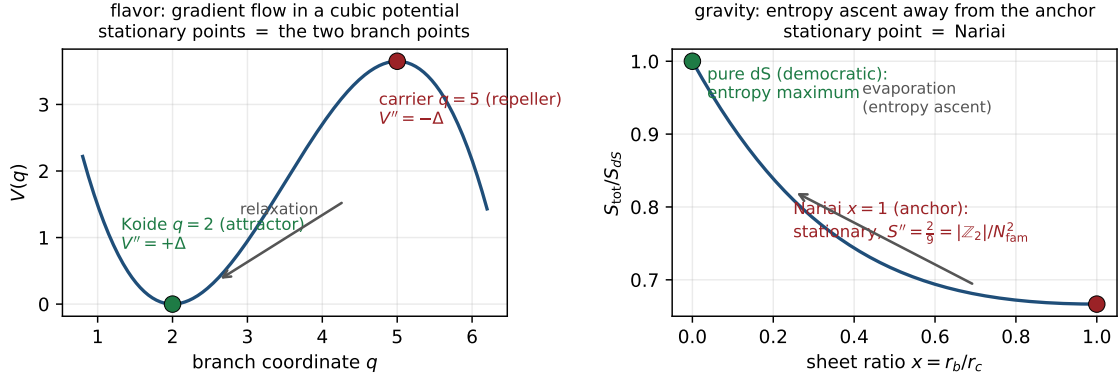


Figure 4: The orientation theorem: the anchor configuration is the stationary repeller of the natural functional in both sectors — flavor (cubic potential, curvatures  $\pm\Delta$ ) and gravity (SdS entropy, curvature  $2/9 = |\mathbb{Z}_2|/N_{\text{fam}}^2$ ). [verification/v102\_seam\_orientation.py]

$2/3 = |\mathbb{Z}_2|/N_{\text{fam}}$ , exponent ratio  $2 = |\mathbb{Z}_2|$ . What is still missing for a full identification is the gravity-side *clock* (the near-Nariai evaporation generator that converts curvature into a rate) — the same class of missing object as the Koide transfer generator of Paper 4: **one clock, two known geometries**. That identification stays [P].

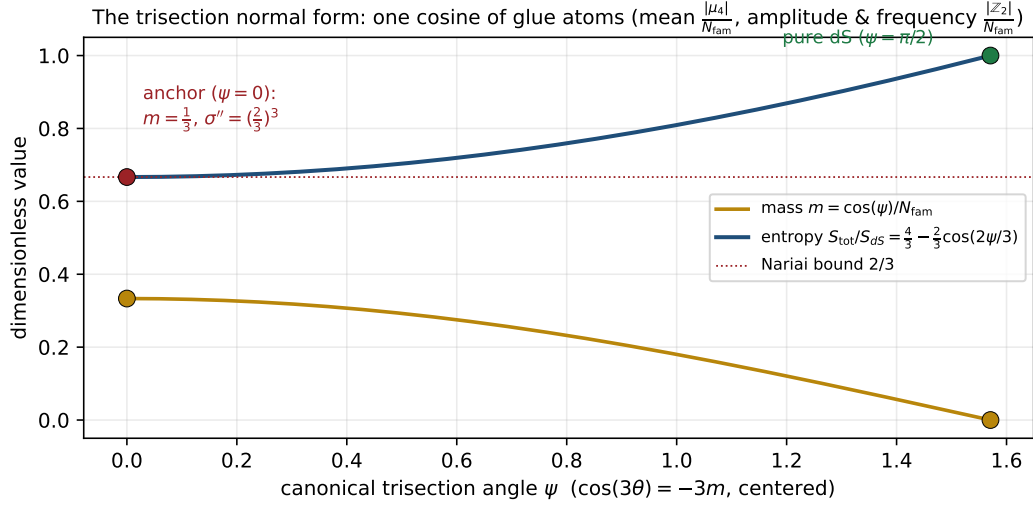


Figure 5: The trisection normal form: in the canonical angle  $\psi$  the mass and the total entropy are pure (co)sines built from glue atoms; the anchor sits at  $\psi = 0$  with canonical curvature  $(2/3)^3$ . [verification/v103\_trisection\_normal\_form.py]

The classical clock speaks anchor — and the honest  $(2/3)$ -test [I]/[P]  
([verification/v104\_nariai\_clock.py])

The clock question of the previous box has a *classical* half, and it is pure GR (Ginsparg–Perry 1983; Bousso–Hawking class): linearize around the Nariai geometry  $dS_2 \times S^2$ . **Static**



**pin [I]:** on the  $dS_2$  static patch the function  $\phi(\rho) = \rho$  satisfies  $\frac{d}{d\rho}[(1 - \Lambda\rho^2)\phi'] = -2\Lambda\phi$  *exactly* — and this static mode *is* the exact SdS two-horizon deformation, so the  $S^2$ -radius modulus mass is pinned internally:  $m^2 = -2\Lambda = -|\mathbb{Z}_2|\Lambda$  (the Laplace-type form of the linearization is the one external GR import, typed honestly). The Euclidean spectrum is the Ginsparg–Perry tower  $(l(l+1) - 2)\Lambda$ : exactly *one* negative mode  $-|\mathbb{Z}_2|\Lambda$ . **The clock [I]:** in Hubble units the homogeneous modes  $\phi \sim e^{\lambda H t}$  obey

$$\chi_{\text{clock}}(\lambda) = \lambda^2 + \lambda - 2 = (\lambda - 1)(\lambda + 2)$$

— **the anchor quadratic:** eigenvalues  $\{1, -2\}$  = the distinct anchor roots, and the Nariai cubic factors as  $(t - 1)^2(t + 2) = (t - 1) \cdot \chi_{\text{clock}}(t)$ . The anchor thus appears a *third* time: as configuration roots, as the curvature base, and as the spectrum of the linearized clock. The horizon split is linear in the canonical angle with coefficient  $1/\sqrt{N_{\text{fam}}}$ , and the entropy deviation grows at exactly  $2H = |\mathbb{Z}_2| \times \text{Hubble}$ . **Honest (2/3)-test [P]:** the classical clock eigenvalues are *integers*, not (2/3)-powers — the (2/3)-powers stay in the functional (curvature  $(2/3)^{N_{\text{fam}}}$ ). The conversion curvature→rate at one loop (quantum (anti-)evaporation) remains external QFT input: the bridge to the flavor rate  $(2/3)^{2N_{\text{fam}}}$  is *not* closed here. The seam variational principle now has exactly *one* missing object: the quantum clock.

The quantum-clock target, made quantitative [I]/[P]  
([verification/v107\\_quantum\\_clock\\_target.py](#))

Instead of guessing the quantum clock, its job description can be pinned exactly. **(1) The classical clock already lives on the cusp ladder [I]:** across the physical Ginsparg–Perry modes ( $l = 0$ :  $(\lambda - 1)(\lambda + 2)$ ;  $l = 1$ :  $\lambda(\lambda + 1)$ ) the decay exponents are

$$\{0, -1, -2\} = -\text{Spec}(V) = -N_{\text{fam}} \times \text{cusp weights}$$

— exactly the grading that carries the transfer operator’s eigenspaces (Papers 2/6); the  $l \geq 2$  modes are damped oscillations with  $\text{Re } \lambda = -\frac{1}{2} = -\delta$  (audit). **(2) The exact target [I]:** the transfer rates  $\{0, \Delta, 6 \ln 3\}$  are *not* weight-linear —  $\text{rate}(2)/\text{rate}(1) = \log_{3/2} 3 = 1 + \log_{3/2} 2 = 2.7095$ , with the exact identity  $(1/3)^6 = ((2/3)^6)^{\log_{3/2} 3}$ : *the quantum clock must bend the weight-linear spectrum by  $\log_{9/4} 3 = 1.35476$  per weight step*. **(3) Coupling landscape [I]:** the dimensionless clock coupling  $\kappa = (c/24\pi)(\Lambda/\bar{M}_{\text{Pl}}^2)$  with  $c = 8$  is  $7.6 \times 10^{-122}$  at the cosmological Nariai (hopeless) and  $\kappa = 1/(3\pi) = 0.106$  at seam curvature — the only regime where an  $O(0.35)$  bend is reachable, and it is borderline non-perturbative: one-loop linear response cannot fix the bend; an  $O(1)$  resummation or an exact seam construction is required. *Status [P]:* R1 is sharpened from “find a clock” to “produce the bend  $\log_{3/2} 3$  on the cusp ladder at coupling  $O(1/3\pi)$ ”. Nothing is fitted; the identification stays open.

## 6 Turnaround radius, gravitational waves, light

**Bound structures.** In a  $\Lambda$ -universe the maximal bound radius is  $r_{\text{ta}}(M) = (GM[2\pi e^{\alpha_*^{-1}}/\bar{M}_{\text{Pl}}]^2)^{1/3}$  — halo, cluster, supercluster and void scales follow from the  $\Lambda$  closure with no new parameter. [P]

**Gravitational waves.** The low-curvature branch is  $R + R^2$ ; the tensor graviton runs on the Lorentz cone and the only extra mode (the scalaron,  $M_{\text{scal}} \approx 3.06 \times 10^{13}$  GeV) is frozen for astrophysical frequencies. Hence

$$v_{\text{GW}} = c \quad (\text{no measurable dispersion}) \quad [\text{P}],$$



| Black-hole mechanics in seam units ( $c_3 = 1/(8\pi)$ ): every coefficient is a compiler atom |                                |                                     |                                                     |
|-----------------------------------------------------------------------------------------------|--------------------------------|-------------------------------------|-----------------------------------------------------|
| quantity                                                                                      | standard form                  | seam form                           | compiler atom                                       |
| Einstein equations                                                                            | $G_{\mu\nu} = 8\pi T_{\mu\nu}$ | $G_{\mu\nu} = T_{\mu\nu}/c_3$       | $c_3 = 1/(8\pi)$ (P1)                               |
| first law                                                                                     | $dM = \frac{\kappa}{8\pi} dA$  | $dM = c_3 \kappa dA$                | seam constant                                       |
| Smarr (Schw.)                                                                                 | $M = \frac{\kappa A}{4\pi}$    | $M = 2c_3 \kappa A$                 |                                                     |
| temperature                                                                                   | $T_H = \frac{\kappa}{2\pi}$    | $T_H = 4c_3 \kappa$                 | $1/(2\pi) = 4c_3$                                   |
| entropy                                                                                       | $S = \frac{A}{4}$              | $S = 2\pi c_3 A$                    | $1/4 = 1/ \mu_4 $                                   |
| Bekenstein bound                                                                              | $S \leq 2\pi ER$               | $S \leq ER/(4c_3)$                  |                                                     |
| Hawking power                                                                                 | $P = \frac{1}{15360\pi M^2}$   | $P = \frac{c_3}{1920 M^2}$          | $1920 =  W(D_5) $                                   |
| lifetime                                                                                      | $\tau = 5120\pi M^3$           | $\tau = 128 g_{\text{car}} M^3/c_3$ | $128 = 2^7$ , 7 = scalaron                          |
| Kerr extremal                                                                                 | $A_{\text{ext}} = 8\pi M^2$    | $A_{\text{ext}} = M^2/c_3$          | $A_{\text{ext}}/A_{\text{Schw}} = 1/ \mathbb{Z}_2 $ |
| area quantum (Hod)                                                                            | $\Delta A = 4\ln 3$            | $\Delta A = \ln(N_{\text{fam}}^4)$  | 81 = disc of the cover                              |
| Nariai bound                                                                                  | $S_N = \frac{2\pi}{\Lambda}$   | $S_N = \frac{2}{3} S_{dS}$          | $2/3 =  \mathbb{Z}_2 /N_{\text{fam}}$ (Koide)       |

Figure 6: Black-hole mechanics in seam units: every classical coefficient is a compiler atom. [\[verification/v101\\_horizon\\_anchor.py\]](#)

a clean falsifier: a genuine  $v_{\text{GW}} \neq c$  would break this gravity closure.

**Light.** TFPT does not predict the SI number  $c$  (definitional); it predicts the *universality of the causal cone* —  $\text{RP} \rightarrow \text{OS reconstruction} \rightarrow$  one Lorentzian cone shared by EM, gravity and massless modes ( $v_\gamma = v_g = c$ , no native photon dispersion). What it *does* compute is not a speed shift but a polarisation rotation, cosmic birefringence

$$\beta_{\text{rad}} = \frac{u}{4\pi} \approx 0.2424^\circ \quad [\mathbf{N}],$$

from the same seed  $u = \phi_0^{\text{ret}}$  that fixes the Cabibbo angle,  $\Omega_b$  and  $\theta_{13}$ .

## 7 Search targets (not claims)

### Audit-level search ansätze [\[A\]](#)

- **Black-hole echoes:** any near-horizon echo amplitude ratio would be a natural transfer-rate target,  $\mathcal{A}_{n+1}/\mathcal{A}_n \lesssim (2/3)^6 \approx 0.0878$ . A search ansatz, no prediction.
- **Quantum recovery:** the Page-curve recovery kernel  $I \sim (2/3)^{6n}$  is a falsifiable shape if a boundary-recovery rate is ever measured.

### Net

All horizon formulas share the one seam normalisation  $1/(2\pi) = 4c_3$ . Black-hole, de Sitter and Unruh temperatures, the Page time, scrambling, the Nariai bound, turnaround radii,  $v_{\text{GW}} = c$  and birefringence are *readouts of one seam constant* — standard physics rewritten in TFPT units, with two genuine compiler fingerprints ( $1920 = |W(D_5)|$ ) in Hawking power,

$|\mu_4| = 4$  in scrambling). No new gravitational physics is claimed; the value of the reframe is that horizons stop being separate modules.

## Appendix: computational verification

The seam-unit identities of this note (the thermal grammar  $\frac{1}{2\pi}=4c_3$ , the power denominator  $1920=|W(D_5)|$ , the de Sitter entropy form, the Page fraction, the scrambling  $|\mu_4|=4$ , and  $\beta_{\text{rad}}=\varphi_0^{\text{ret}}/4\pi=0.2424^\circ$ ) are re-derived and machine-checked in `verification/v8_horizon.py`. Run `python verification/v8_horizon.py` (needs `mpmath`); it imports the same  $c_3=1/(8\pi)$  primitive as the rest of the suite. The search-target ansätze remain **[A]** and are deliberately not part of the pass/fail checks.